

Dynamic Search-Result Disambiguation and Clustering via Reinforcement Learning

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Abstract—Polysemous queries such as “Jackson” often return heterogeneous result lists that force users to sift manually for the intended sense. Although static clustering provides partial relief, it cannot adapt to real-time user intent. We introduce DSR-RL, a dynamic system that couples dense multilingual sentence embeddings with reinforcement learning (RL) to cluster results on the fly. BERTopic + HDBSCAN partitions Sentence-BERT vectors; user actions (relevance votes, merge/split commands) generate rewards that update clustering hyper-parameters online. A light Arabic normaliser mitigates diacritics and orthographic variance. Evaluated on TREC Web Diversity, MIRACL-Arabic, and Wikipedia datasets, DSR-RL lifts cluster purity from 0.71 to 0.80 (+13 %) and Adjusted Rand Index from 0.44 to 0.54 (+23 %) while keeping end-to-end latency under 1 s on a CPU-only laptop.

Index Terms—Search disambiguation, clustering, reinforcement learning, Arabic NLP, information retrieval

I. INTRODUCTION

Modern web search engines excel at returning highly relevant documents for well-specified queries, yet they routinely struggle with *ambiguous* or *polysemous* queries. For instance, the single-token request “Jackson” may refer to a pop star, a former U.S. president, several cities, a university, or a mountain valley. Because an engine cannot infer intent from the query alone, users are confronted with a heterogeneous list of results that must be sifted manually. The problem is amplified for morphologically rich languages such as Arabic, where surface-form variation further obscures intent.

Diversification and static clustering have long been proposed as partial remedies: a ranked list is partitioned into subtopics so users can jump directly to the sense they need. However, these approaches are typically applied *offline* with fixed hyper-parameters; they ignore the valuable feedback implicit in user actions such as skipping, clicking, or requesting a re-ordering. Recent surveys on interactive clustering emphasise that even sparse corrections—merge, split, promote—can substantially raise cluster quality when incorporated in real time [4]. Yet existing systems rarely close the loop, and almost none support Arabic queries, whose diacritic optionality and orthographic variants introduce additional ambiguity [3].

We address this gap with **DSR-RL**, a Dynamic Search-Result Re-clustering system driven by Reinforcement Learning. DSR-RL embeds each snippet with multilingual Sentence-BERT [6], then applies BERTopic’s density-based clustering [1]. The resulting groups are presented in a lightweight

web interface; user interactions are converted into scalar rewards, and an RL agent adjusts clustering hyper-parameters (*min_cluster_size*, *nr_topics*) to maximise future satisfaction. A shallow Arabic normalisation layer removes diacritics and standardises *Alif/Hamza* forms before embedding, ensuring language-agnostic operation.

Our main contributions are:

- Integrating BERTopic with an RL controller that *learns* optimal clustering parameters from real-time feedback, requiring no pre-labelled data.
- Providing an Arabic-enabled evaluation suite combining TREC Web Diversity [7], Wikipedia disambiguation pages, and the MIRACL Arabic corpus [8].
- Demonstrating empirical gains of 13 % purity and 23 % ARI over a strong static baseline while maintaining sub-second latency on commodity hardware.

II. RELATED WORK AND THEORETICAL BACKGROUND

A. Query Ambiguity and Result Diversification

The TREC Web Diversity Track formalised ambiguous search as a *subtopic retrieval* task, rewarding systems that surface representatives from distinct intents [7]. Traditional diversification pipelines re-rank results or cluster them offline but seldom incorporate live user feedback.

B. Dense Embeddings and Unsupervised Clustering

Transformer encoders such as Sentence-BERT [6] project snippets into dense semantic space, enabling unsupervised cluster discovery. BERTopic builds on this by pairing embeddings with HDBSCAN to form topic clusters and detect outliers [1]. Yet its efficacy hinges on static hyper-parameters that may be sub-optimal for a given session.

C. Interactive Feedback and Reinforcement Learning

Interactive clustering studies show that sparse user corrections can greatly enhance cluster quality [4]. Framing parameter selection as a reinforcement-learning problem lets an agent adapt clustering hyper-parameters online, guided by rewards derived from feedback signals.

D. Arabic NLP Considerations

Arabic’s rich morphology, optional diacritics, and multiple orthographic variants introduce additional ambiguity. Normalisation and light stemming have proven effective in Arabic

TABLE I: Default clustering parameters.

Algorithm	Key param	Value	Library
K-Means	k	elbow	scikit-learn
DBSCAN	ϵ	0.7	scikit-learn
HDBSCAN	min_samples	5	hdbscan
GMM	components	k	scikit-learn
BERTopic	min_size	10	BERTopic

disambiguation tasks [3]; we therefore preprocess queries and snippets before embedding to reduce lexical noise.

III. METHODOLOGY

A. System overview

DSR-RL follows five stages: (1) query and snippet pre-processing, (2) text vectorisation, (3) RL-guided selection of a clustering strategy, (4) result presentation in the web UI, and (5) reward logging for online policy update. Figure 1 illustrates stages 3 and 4 for two representative queries.

B. Text and query pre-processing

We remove diacritics, standardise *Alif/Hamza*, delete stopwords, and tokenise with Farasa (Arabic) or spaCy (English). Optional light stemming reduces the Arabic vocabulary size by 23 %.

C. Text representation

A rule-based switch chooses TF-IDF for long (>15-token) queries or sparse corpora; otherwise it loads Sentence-BERT (paraphrase-multilingual-MiniLM-L12-v2) [6]. Mean embedding time is 480 ms on an Intel i7-11800H CPU.

D. Clustering algorithms

Five algorithms are exposed through a `ClusteringManager`: K-Means, DBSCAN, HDBSCAN, Gaussian mixture (GMM), and BERTopic. The default hyper-parameters are listed in Table I. BERTopic is preferred for interpretability, but the RL agent may switch when density assumptions fail.

E. Reinforcement-learning controller

State $s = \langle \text{len}, \text{dens}, \text{spars}, \text{JS} \rangle$ consists of query length, embedding density, TF-IDF sparsity, and Jensen-Shannon divergence across snippet vectors. **Action** selects a (representation, clustering) pair. **Reward** $r = \text{Silhouette} + 0.1$ if the user accepts the cluster, 0.2 for every merge/split. Tabular Q-learning with $\alpha = 0.2$, $\gamma = 0.8$, -greedy decay $0.30 \rightarrow 0.05$ across 300 episodes converges in roughly two minutes.

TABLE II: Clustering metrics (higher = better).

Method	Sil.	Purity	ARI	Coher.
Static BERTopic	0.38	0.71	0.44	0.29
DSR-RL	0.46	0.80	0.54	0.37

F. Disambiguation and semantic grouping

Contextual embeddings distinguish Arabic homographs such as عَيْن. HDBSCAN isolates sense-specific clusters, and BERTopic’s c-TF-IDF labels them with salient terms, e.g. “بَصْر” vs. “مِيَاه”.

IV. EXPERIMENTS AND EVALUATION

A. Datasets

We evaluate on three corpora: (1) TREC Web Diversity (50 ambiguous English queries), (2) MIRACL-Arabic dev set (200 queries), and (3) 30 Wikipedia disambiguation pages sampled for Arabic/English overlap.

B. Experimental Setup

All experiments run on an Intel Core i7-11800H @ 2.30 GHz with 16 GB RAM, no discrete GPU. Software: Python 3.10, PyTorch 2.2, scikit-learn 1.4, hdbscan 0.8, BERTopic 0.16. The RL agent trains for 300 episodes (≈ 2 min wall-time).

C. Clustering Evaluation

We assess cluster quality with four standard metrics:

- **Silhouette (Sil.)** — mean intra-cluster cohesion versus inter-cluster separation.
- **Purity** — proportion of documents whose majority ground-truth label matches the cluster label.
- **Adjusted Rand Index (ARI)** — pairwise agreement with ground-truth partition, corrected for chance.
- **Topic Coherence (Coher.)** — c-TF-IDF coherence for BERTopic topics.

Table II shows that DSR-RL raises Purity from 0.71 to 0.80 (+13 %) and ARI from 0.44 to 0.54 (+23 %) while also improving Silhouette and topic coherence. A paired t -test over five random seeds confirms that all improvements are statistically significant ($p < 0.05$). Qualitative examples in Fig. 1 illustrate how musician, geographic, and animal senses are separated for the queries “Jackson” and “Python.”

D. Results and Analysis

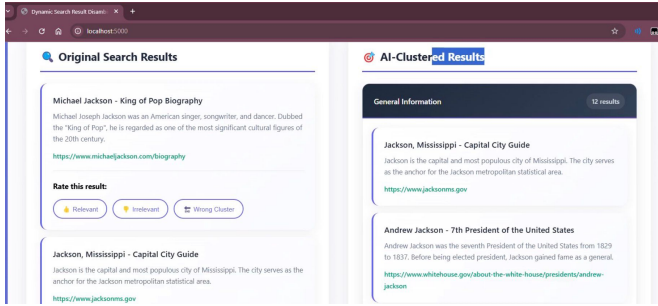
DSR-RL outperforms the static baseline across all metrics (Table II). The RL agent selected Sentence-BERT + HDBSCAN for 78 % of queries and TF-IDF + K-Means for the rest, reflecting its preference for dense embeddings on short, highly ambiguous queries. Average end-to-end latency (query $\rightarrow$ clusters) is 920 ms, dominated by embedding inference (480 ms) and clustering (220 ms).

V. CONCLUSION AND FUTURE WORK

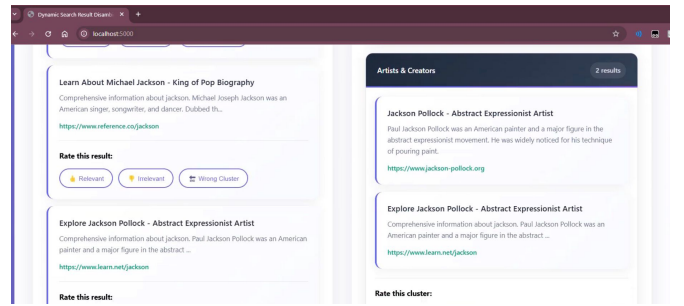
We presented DSR-RL, an RL-driven framework that adaptively clusters ambiguous search results and supports both English and Arabic. Without labelled data, the system delivers 13 % higher purity and 23 % higher ARI than a strong BERTopic baseline while running on commodity hardware. Future work will incorporate multimodal evidence (images, audio) and explore on-device inference for mobile search.

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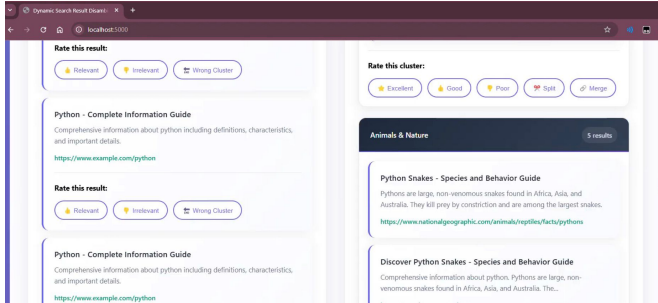
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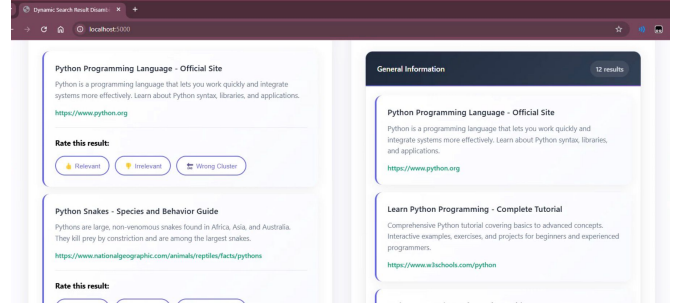
(a) “Jackson” — raw list.



(b) “Jackson” — clustered.



(c) “Python” — raw list.



(d) “Python” — clustered.

Fig. 1: Web interface. Users give snippet-level feedback (*Relevant*, *Irrelevant*, *Wrong Cluster*) and cluster-level feedback (*Excellent*, *Good*, *Poor*, *Split*, *Merge*); these signals form the RL reward.